DATA ANALYSIS PROJECT REPORT

Project Title	Bangalore Dining Dynamics: Analyzing Ratings, Cost, and Popularity
Student Name	Stephen Argho Das
Student ID	23-54439-3
Course / Section	Course: Programming in Python, Section: C
Instructor	Md. Tanzeem Rahat
Date of Submission	02/05/2026

1. Executive Summary

This project investigates the dynamics of the restaurant industry in Bangalore using a dataset of over 50,000 Zomato listings. The primary goal was to understand how restaurant characteristics—such as type, location, and cost—affect customer ratings and popularity (votes). Using Pandas for data cleaning, NumPy for custom statistical scoring, and Matplotlib for visual analysis, the dataset was processed to handle missing values and inconsistent formatting. Key findings reveal that "Drinks & Nightlife" and "Pubs and Bars" yield the highest average ratings, whereas higher costs do not guarantee higher ratings beyond the 4.0 mark. Additionally, extreme outliers exist in popularity, indicating that a small percentage of restaurants capture the vast majority of consumer attention.

2. Introduction and Project Objectives

Item	Your Content
Dataset name	Zomato Bangalore Restaurants
Dataset source	https://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurants
Main project goal	[State the overall goal in 1–2 sentences]
Research Question 1	How do restaurant ratings vary across different listed types (e.g., dining, delivery, nightlife) in Bangalore?
Research Question 2	What is the relationship between restaurant cost, customer ratings, and overall popularity (votes)?
Research Question 3 (optional)	What characteristics define highly popular restaurants, and are there significant outliers in terms of votes or pricing?

3. Dataset Description

The Dataset consists of 51717 rows and 17 columns. The columns represents the variables: url, address, name, online_order, book_table, rate, votes, phone, location, rest_type, dish_liked, cuisines, approx_cost(for two people), reviews_list, menu_item, listed_in(type), listed_in(city)

The basic idea of analyzing the Zomato dataset is to get a fair idea about the factors affecting the establishment of different types of restaurants at different places in Bengaluru, aggregate rating of each restaurant, Bengaluru being one such city has more than 12,000 restaurants with restaurants serving dishes from all over the world. The data is accurate to that available on the zomato website until 15 March 2019.

3.1 Basic Dataset Information

Property	Value
Number of rows	51717 (before cleaning)
Number of columns	17 (before cleaning)
Data types present	Numerical, Categorical, Text
Target or key variable(s)	rate, votes, approx_cost

Potential issues observed at first glance	Missing values, Outliers, Inconsistent Row Entry's

3.2 Initial Observations

Upon initial inspection, the dataset is highly robust but requires heavy text-parsing to extract usable numbers. There is a clear imbalance in locations, with hubs like BTM and Koramangala having vastly more entries than others. The votes column appears highly skewed, suggesting a few very popular places dominate the app.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
rate column formatted as string (e.g., "4.1/5", "NEW")	Removed "/5", replaced "NEW" and "-" with NaN, converted to float.	Statistical analysis (mean, correlation) requires clean numerical data.	Section 4, Step 2
approx_cost contained commas as strings (e.g., "1,200")	Removed commas using string replacement and converted to float.	Necessary for computing average costs and relationship analysis with ratings.	Section 4., Step 3
Redundant/heavy text columns (url, reviews_list, phone)	Dropped these columns entirely.	These columns consume high memory and are not required for aggregate statistical analysis.	Section 4, Step 1
Missing vital metrics	Dropped rows containing Nan in rate or cost columns.	Imputing ratings or costs could introduce heavy bias into a relationship analysis.	Section 4., Step 4

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
cost_per_person	Divided approx_cost(for two people) by 2	Counted the number of items in the comma-separated cuisines string.
cuisine_count	Counted the number of items in the comma-separated cuisines string.	Helps analyze if offering a wider variety of cuisines impacts ratings or popularity.
popularity_tier	Used np.select to evaluate the numerical votes column and group them into three distinct categorical bins based on thresholds: 'Low' (<50), 'Medium' (50 to 499), and 'High' (≥500).	The original votes column is highly skewed with extreme outliers (0 to 16,000+). Converting this into categorical tiers makes it much easier to perform subgroup comparisons and creates cleaner, more readable visualisations.

6. Analysis Methodology

- Pandas was used for data ingestion, string manipulation (cleaning numeric columns), handling missing values, creating derived features, and grouping data by location and restaurant type.

- NumPy was used for fast array operations, handling NaN values safely (np.nanmean, np.nanstd), conditional logic (np.where), and generating standardized Z-scores.
- Matplotlib was used to create visual evidence directly addressing the Research Questions, utilizing bar charts, scatter plots with color mapping, and boxplots for distribution analysis.

7. Analysis and Visualizations

7.1 Research Question 1

How do restaurant ratings vary across different listed types in Bangalore?

To answer this, I grouped the data by rest_type_group (listed_in type) and calculated the mean rating.

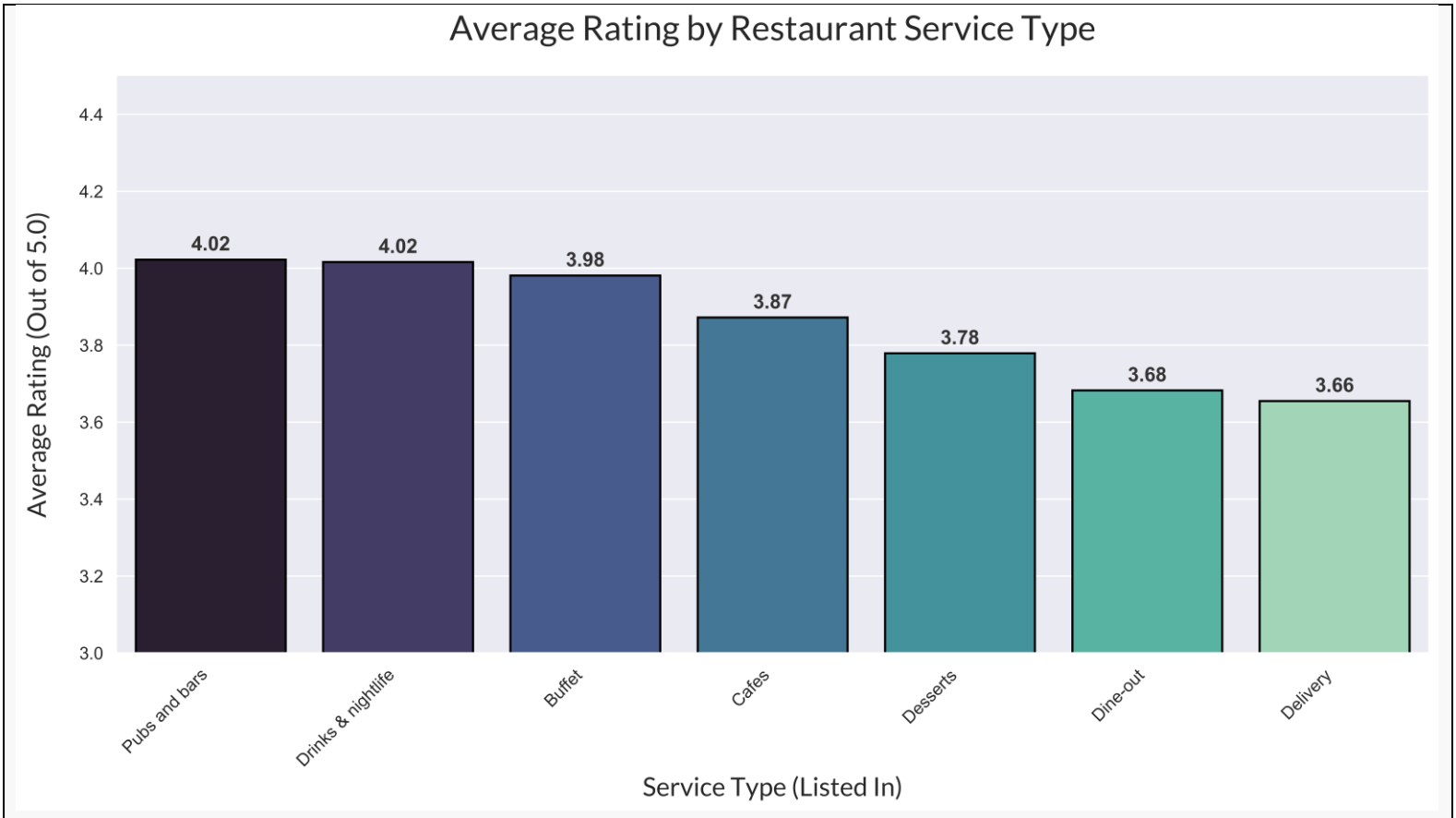

Figure 1. Average Rating by Restaurant Service Types

Interpretation: The analysis reveals a distinct difference in how customers rate different types of dining experiences. Restaurants categorized under entertainment and nightlife, specifically "Drinks & nightlife" and "Pubs and bars," yield the highest average ratings (approximately 4.0 or higher). In contrast, standard utility services like "Delivery" and "Dine-out" have noticeably lower average ratings (around 3.6 to 3.7). This suggests that customers tend to be more generous or satisfied with their ratings when paying for nightlife and entertainment experiences compared to standard food delivery or everyday dining.

7.2 Research Question 2

What is the relationship between restaurant cost, customer ratings, and overall popularity (votes)?

I utilized a scatter plot combining three variables: Cost (x-axis), Rating (y-axis), and Popularity/Votes (color map using a logarithmic scale to account for skewness).

Correlation between Cost and Rating: 0.39

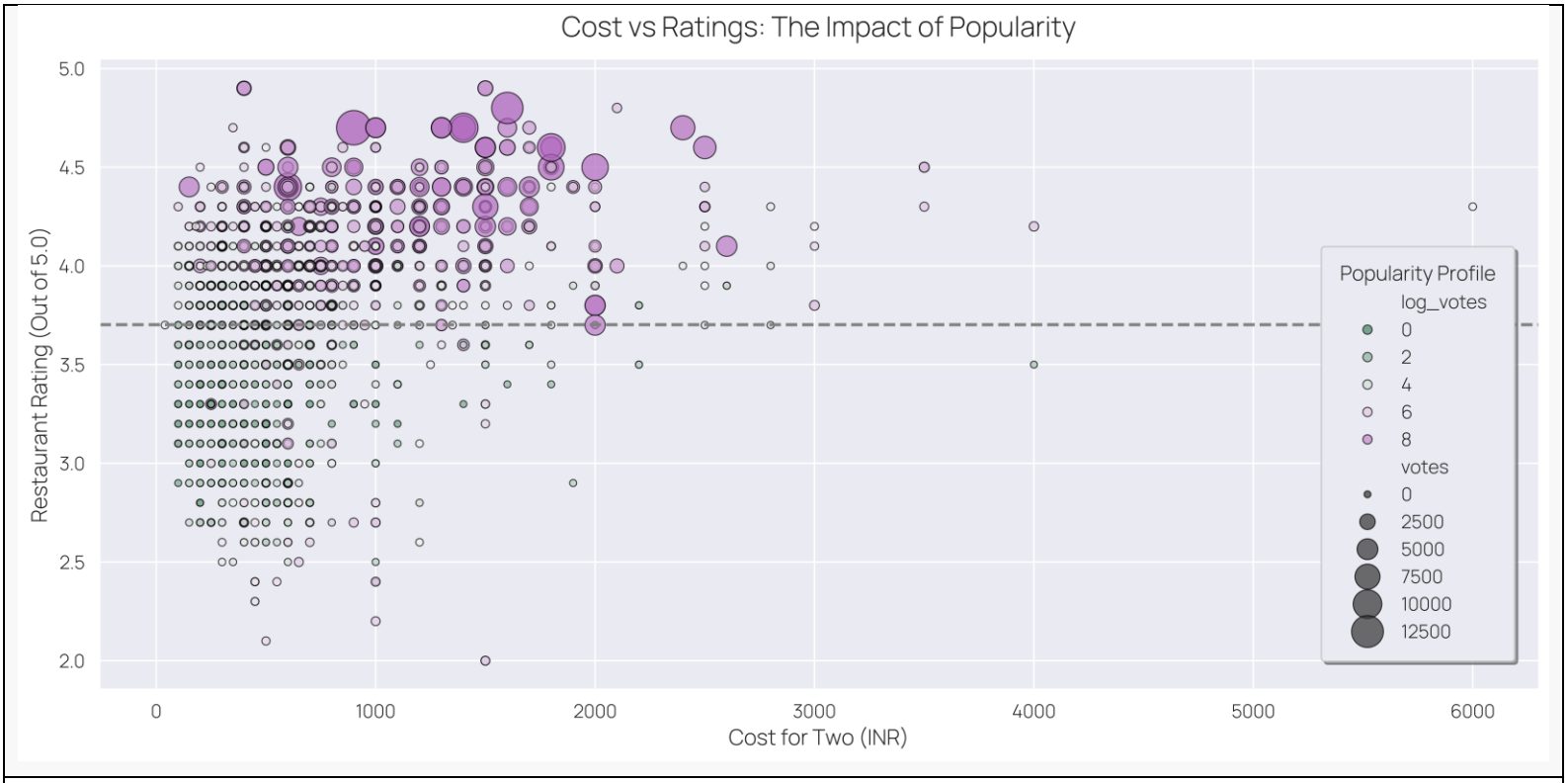

Figure 2. Cost and Ratings impact on Popularity

Interpretation: There is a positive, albeit imperfect, correlation (0.39) between cost and ratings. Extremely low-cost restaurants rarely break the 4.5 rating mark, while higher-cost restaurants generally sit above the average rating line. However, cost is not a perfect predictor of quality; beyond ₹3000 (for two people), ratings tend to plateau around 4.2–4.5 regardless of price increases. When factoring in popularity (votes), the data shows that luxury is not the driver of engagement. The absolute most popular restaurants are heavily clustered in the mid-range cost tier (₹500 to ₹1500 for two). This proves that maximum customer engagement and popularity occur at moderate price points, rather than at luxury establishments.

7.3 Research Question 3

What characteristics define highly popular restaurants, and are there any extreme outliers in terms of votes or pricing?

I created boxplots to visualize the distribution and detect statistical outliers in both the votes and cost_for_two columns.

99th Percentile of Votes: 4313 and 99th Percentile of Cost for Two: ₹2500 (Anything above is extreme)

Figure 3. Finding the characteristics popular restaurant and extreme outlier in terms of vote and pricing

Interpretation: Highly popular restaurants are defined by providing high consumer value (Bang for the Buck), where mid-tier cafes frequently outscore luxury hotels when standardizing ratings against cost. The data reveals extreme statistical outliers and a massive right-skew in both popularity and pricing:

Popularity (Votes): There is a 'winner-takes-all' dynamic in the market. While the vast majority of restaurants have a very low median vote count (fewer than 100 votes), a tiny fraction of restaurants dominates the platform's traffic, acting as extreme outliers that pull in tens of thousands of votes (up to 16,000+).

Pricing: A similar outlier pattern exists in the pricing model. The 99th percentile for cost is ₹2500, meaning luxury dining establishments (charging up to ₹6000) act as heavy mathematical outliers compared to the general Bangalore restaurant market.

8. NumPy-Based Custom Computation

Component	Description
Computation used	value_score via Z-score Standardization
Purpose	To objectively measure which restaurants offer the best "Bang for the Buck" by mathematically comparing two different scales (Ratings 1-5 vs Cost 100-6000).
Formula or logic	$Z_{rate} = \frac{(rate - mean(rate))}{std(rate)}$, $Z_{cost} = \frac{(cost - mean(cost))}{std(cost)}$, $\text{value_score} = \text{np.where}(\text{valid_data}, Z_{rate} - Z_{cost}, \text{NaN})$
Result / takeaway	Restaurants with positive value scores offer higher-than-average ratings at lower-than-average costs. This proved that mid-tier cafes often outscore luxury hotels in raw consumer value.

9. Key Findings

1. Nightlife Commands Higher Ratings: Restaurants categorized under "Pubs and Bars" or "Drinks & Nightlife" consistently receive higher average ratings (approx 4.0+) than standard delivery places (approx 3.6).

2. Mid-Range Dominates Popularity: The scatter plot analysis reveals that the highest volume of user votes occurs in the mid-range cost bracket (₹500 to ₹1500 for two). Luxury restaurants, while rated highly, have significantly lower user engagement volume.

3. Extreme Market Concentration (Outliers): The popularity (votes) metric is heavily right-skewed. A vast majority of restaurants have fewer than 100 votes, while a few outliers pull in tens of thousands, indicating extreme market concentration.

4. Cost is Not a Perfect Predictor of Quality: While very cheap restaurants rarely achieve 4.5+ ratings, spending over ₹3000 for two does not guarantee a perfect score, as ratings plateau around 4.2-4.5 regardless of how much more expensive the food gets.

5. Geographic Concentration: As shown in the density analysis, BTM Layout and Koramangala are the absolute epicenters of Bangalore's restaurant scene, hosting thousands more businesses than other neighborhoods.

10. Limitations

The primary limitation of this dataset is sample bias. Because this is Zomato app data, it only reflects the opinions of a specific digitally-active demographic, not the entire population of Bangalore. Furthermore, the handling of missing ratings by dropping rows removes newly opened restaurants from our analysis, meaning our findings only apply to established businesses. Finally, the positive relationship between cost and ratings indicates correlation, not causation; more expensive restaurants likely have higher ratings due to better ambiance and ingredients, not simply because they charge more.

11. Conclusion

This analysis successfully mapped the dynamics of Bangalore's restaurant industry. The data suggests that for a new restaurant to achieve high popularity, targeting the mid-range price segment in areas like Koramangala or BTM layout provides the highest user engagement. Furthermore, while high prices correlate slightly with better ratings, consumer value (as calculated by our NumPy metric) remains a driving factor for widespread popularity. Future work could include natural language processing (NLP) on the reviews_list to see exactly *why* customers rate nightlife higher than delivery.

12. References / Dataset Source

Dataset Source: Kaggle (https://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurants**)**

13. Appendix (Optional)

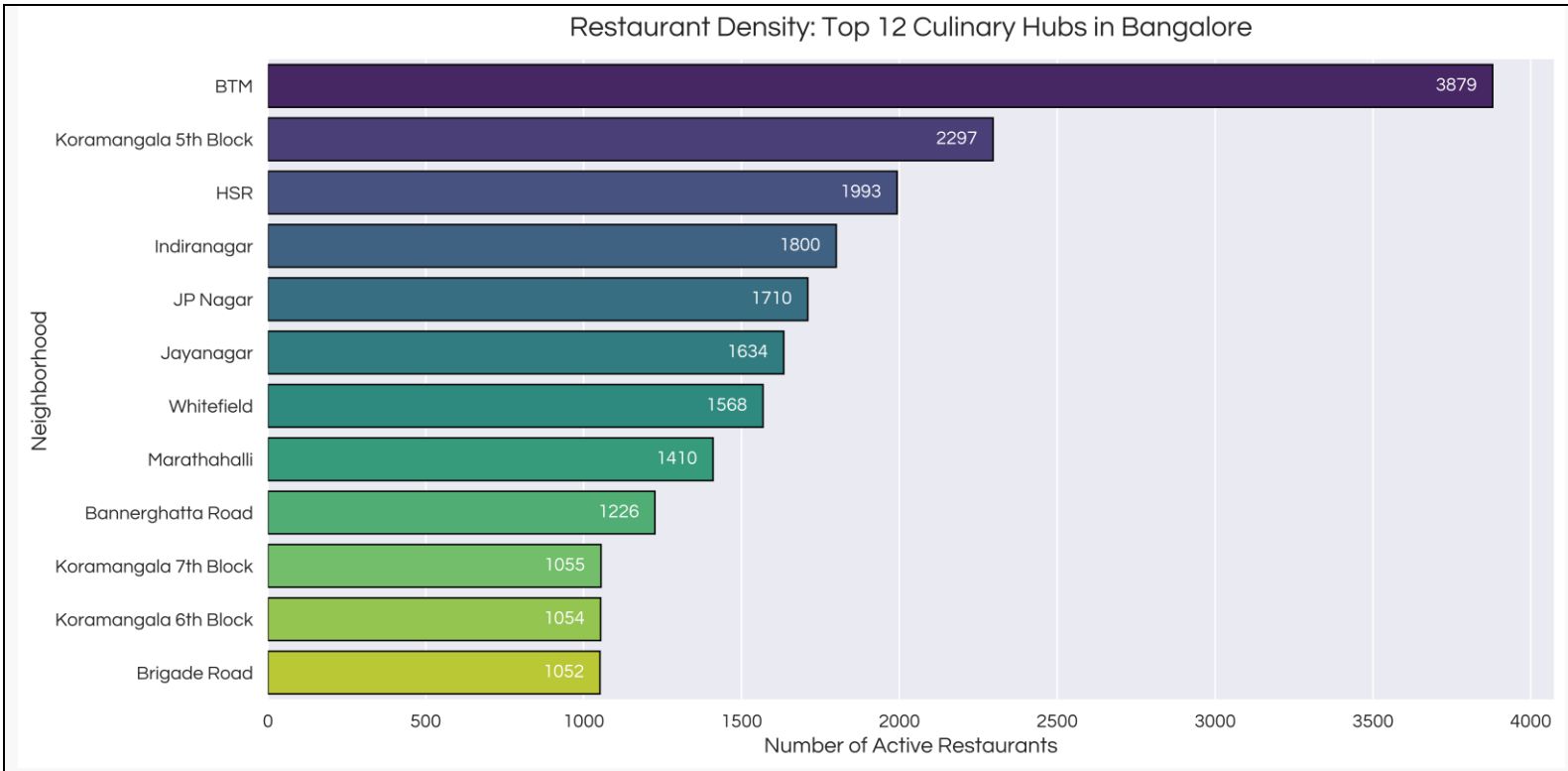

Figure 4. Restaurant Density by location across Bangalore

Final Submission Checklist

- ✓ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ✓ I attached the project report as a separate file.
- ✓ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusion.
- ✓ My code uses Pandas, NumPy, and Matplotlib meaningfully.
- ✓ All bracketed placeholder text and helper notes were replaced or deleted before submission.